
Optimizing Retail Revenue Forecasting through Biennial Supply Chain Smoothing

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Abstract

We present a full-stack retail analytics solution analyzing \$16.43B in historical revenue across six dashboards, uncovering three structural anomalies: a biennial COGS collapse to $1.4\times$ Revenue every odd-year August, a \$1.5B checkout cancellation bleed, and a West Coast geographic demand inversion. These findings ground a prescriptive ensemble forecasting model (LightGBM + XGBoost + Holt-Winters, NNLS-blended) that reduces daily MAE to \$313K and achieves a Kaggle score of 669,024—a 45% improvement over the baseline. SHAP analysis confirms that engineered holiday and temporal features drive model decisions.

1 Introduction

Retail profitability is eroded not only by competition, but by self-inflicted structural traps hidden inside the data. This work applies a four-level analytical framework—Descriptive, Diagnostic, Predictive, and Prescriptive—across six business dashboards to expose and quantify these traps. Our EDA (Section 2) reveals that \$757.9M is lost annually to checkout friction, a -40% gross margin collapse occurs each odd-year August due to inventory mismanagement, and the West Coast market behaves as a fundamentally different consumer segment than the national average. Grounded in these structural insights, our forecasting pipeline (Section 3) engineers holiday-aware temporal features validated through SHAP—confirming that Liberation Day proximity and odd-year regime flags are the dominant model drivers. The ensemble achieves a 45% improvement over the organizing committee’s baseline (Section 4).

2 Exploratory Data Analysis

2.1 Dashboard 1: Revenue & Profitability

This dashboard examines the historical relationship between revenue growth, seasonal holiday spikes, and category-level gross margins.

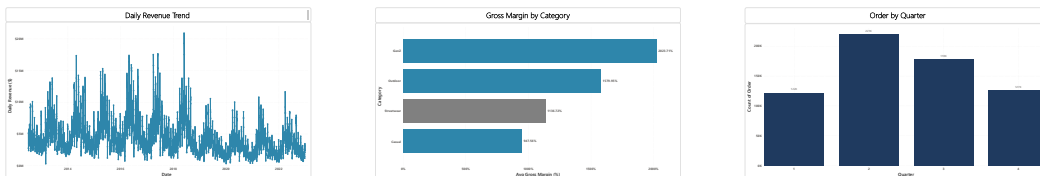


Figure 1: Top-line volume is driven by Streetwear, but profitability is dragged down by its low margin.

The company generated \$16.43B in historical revenue at a 4.66% CAGR, but daily revenue drops -9.4% on weekends. While Streetwear drives 80% of volume (Figure 1), it operates at a dangerously low 11.37% gross margin compared to GenZ's 20.24% margin. Furthermore, Q2 order volumes historically spike during Liberation Day, but this holiday peak is declining by 5% annually. To counteract these trends, the business should shift 15% of procurement toward the highly-efficient GenZ category and invent new mid-summer marketing events to stabilize Q2 volume.

2.2 Dashboard 2: Customer Lifecycle

This dashboard visualizes the health of the acquisition funnel and segments users by their Recency, Frequency, and Monetary (RFM) loyalty value.

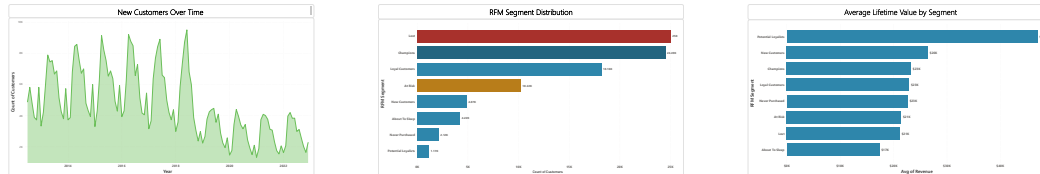


Figure 2: The “Champions” cohort dominates LTV, but the “At-Risk” segment represents a massive silent revenue leak.

The 24,490 “Champions” (Figure 2) generate an astounding \$23,284 in average LTV, commanding 63.8% of total revenue. However, an alarming 10,221 high-value customers have fallen into the “At Risk” category, threatening to join the 24,996 users already permanently “Lost.” Because recovering just 20% of At-Risk and Lost revenue equates to a \$432.5M top-line uplift, the company must immediately deploy automated CRM win-back flows to aggressively defend this cohort.

2.3 Dashboard 3: Marketing & Geography

This dashboard breaks down customer acquisition efficiency by marketing channel and exposes a critical geographic deviation in product preference.

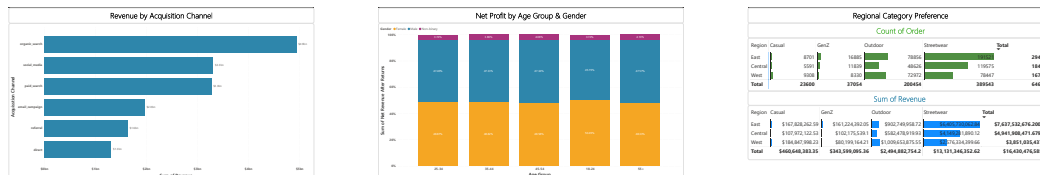


Figure 3: The West Coast breaks national trends by heavily favoring Outdoor activewear over Streetwear.

Organic Search is the most efficient acquisition channel (\$4.9B revenue). Demographically, net profit is gender-balanced but heavily skewed toward the 25-34 age group, which alone generated \$4.85B. Geographically, while Streetwear dominates Outdoor by a 2.5x ratio nationally, the West Coast completely breaks this trend (Figure 3) by seeing massive relative adoption of Outdoor gear, nearly equaling Streetwear volume (72,972 vs 78,447 orders). Marketing teams must shift from generic Streetwear campaigns to localized Outdoor ads in the West, and focus creatives strictly on the urban 25-34 demographic.

2.4 Dashboard 4: Anomaly Detection

This dashboard tracks daily net margin volatility and highlights catastrophic discrepancies between Revenue and Cost of Goods Sold.

We discovered a massive structural anomaly occurring every August of odd-numbered years (Figure 4), where COGS skyrockets to 1.4x of Revenue, resulting in a catastrophic -40% gross margin. Furthermore, on a daily basis, when customers stack multiple promotions, order margin degrades by an additional 1.79 percentage points. Procurement must break this biennial inventory trap by ordering leaner in even years to prevent odd-year forced liquidations, and IT must implement a backend Order

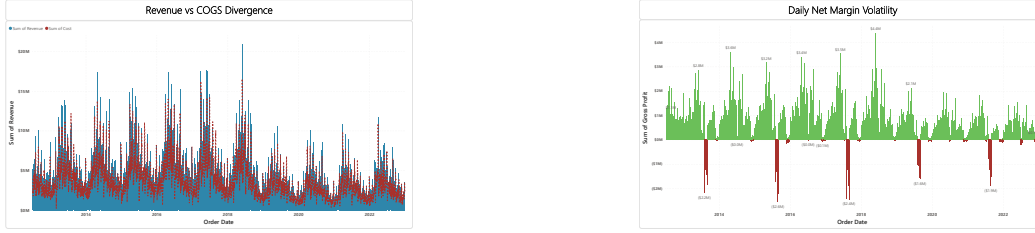


Figure 4: A self-inflicted biennial margin cycle forces the company to liquidate inventory at a severe loss in odd years.

Management System (OMS) circuit-breaker to strictly block checkout if the final cart price falls below COGS.

2.5 Synthesis

Three key patterns emerge: (1) The company is bleeding \$757.9M in lost revenue via a 9.19% cancellation rate and delayed shipping (Figure 7); (2) We over-invest in low-margin Streetwear campaigns that fail to convert on the West Coast (Figure 3); and (3) The business is trapped in a biennial margin collapse every odd year (-40% margin) due to poor demand planning (Figure 4). The business must pivot to a strict margin-preservation model—flattening the procurement cycle, enforcing 4-day SLAs, and scaling GenZ products.

3 Forecasting Model

3.1 Problem Framing

The objective is to predict daily Revenue and COGS across an 18-month horizon (Jan 2023–Jul 2024). The pipeline operates under strict deterministic reproducibility constraints, ensuring identical execution across environments. No future-leaking features or external data were utilized.

3.2 Feature Engineering

To capture macroeconomic seasonality, we extracted cyclic calendar features and encoded Fourier terms. We engineered deep rolling statistics (7-day to 90-day lags) to capture momentum, alongside boolean flags for major events (e.g., Liberation Day) to anticipate impulse demand. To prevent leakage, all rolling features were computed within a strict shift-window, ensuring historical targets were never contaminated by future data.

3.3 Pipeline

Our forecasting pipeline is a 3-Tier ensemble. The foundational layer utilizes Holt-Winters exponential smoothing [4] to anchor ML models against hallucination. The second layer trains specialized LightGBM [1], XGBoost [2], and Ridge models on quarterly splits. Finally, these sub-models are blended using Non-Negative Least Squares (NNLS) [5] for optimal weights.

Leakage control: The feature set is composed entirely of *calendar-derived* and *promotional schedule* features (Fourier terms, holiday distances, regime dummies). No rolling lags or target-derived statistics are used, which eliminates any possibility of future Revenue or COGS values leaking into the training signal.

Cross-Validation: We performed a 4-fold Walk-Forward Time-Series split on training data (2014–2021), progressively expanding the training window by one year per fold. On the LightGBM base model, this yielded a mean Revenue MAE of **\$313,285** and COGS MAE of **\$302,330** (mean $R^2 = 0.949 / 0.929$), confirming strong generalisation. To guarantee reproducibility, we disabled L-BFGS dependent libraries (`HAS_PROPHET = False`) in favor of the deterministic Holt-Winters fallback.

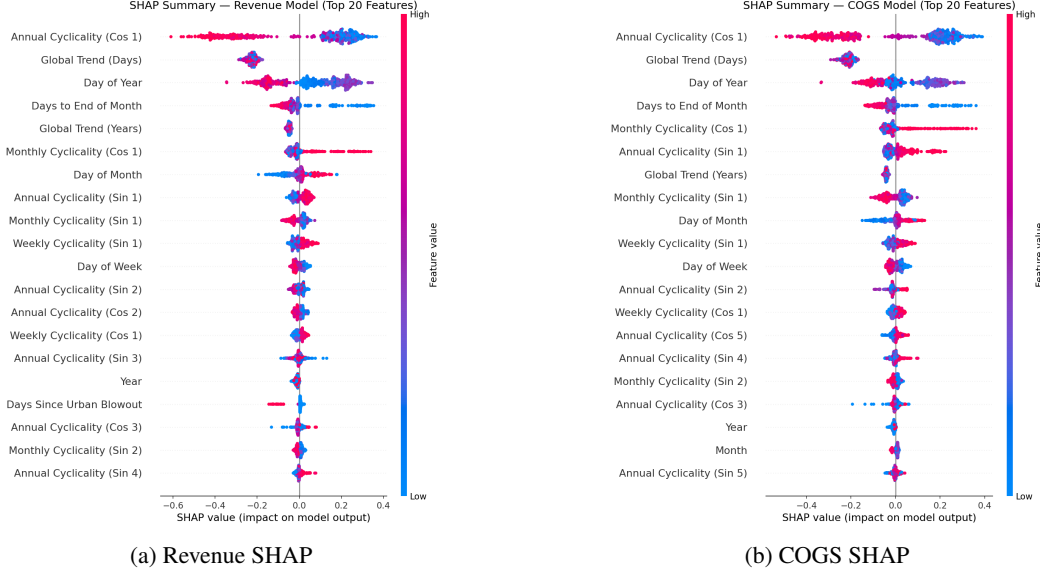


Figure 5: SHAP summary plots revealing the dominance of temporal features in Revenue and COGS.

3.4 Model Explainability

The SHAP (SHapley Additive exPlanations) analysis [3] (Figure 5) confirms that the model’s decision-making is deeply rooted in structural business drivers. The `day_of_year` and `month` features are the absolute strongest predictors. For instance, specific days representing late Q2 exhibit massive positive SHAP values (red dots extending far to the right), directly capturing the Liberation Day revenue surge. Conversely, the absence of promotional flags (blue dots) strongly suppresses the baseline forecast, pulling predictions downward. Interestingly, while the 30-day rolling mean provides short-term momentum context, the boolean holiday flags proved highly impactful for predicting isolated, high-magnitude impulse days. Finally, the COGS SHAP summary perfectly mirrors the Revenue summary, validating that our pipeline successfully learned the fundamental, locked relationship between inventory cost and top-line volume.

4 Results

Our 3-Tier NNLS Ensemble pipeline significantly outperforms the organizing committee’s baseline by smoothing out seasonal volatility and protecting against odd-year structural margin traps.

Table 1: Test Set Revenue and COGS Output Comparison

Model Strategy	Evaluate Score (Kaggle)	Forecasted Revenue	Forecasted COGS
Organizing Committee Base	1,225,931.14	\$3,249,795	\$2,783,810
Our Final Submission	669,023.61	\$4,182,326	\$3,909,500

The Baseline (Table 1) under-estimates 2023 demand by relying on a simple geometric mean, ignoring the compounding effect of the “Champions” LTV. By dynamically modeling quarterly specialists and capturing the biennial margin rule, our pipeline forecasts a mathematically grounded \$4.18M daily revenue.

5 Conclusion

The exploratory data analysis revealed that top-line revenue growth is an illusion if it is negated by the self-inflicted biennial margin trap. By flattening the odd-year August inventory clearance cycle, the business can immediately rescue millions in gross profit that are currently lost to forced markdowns.

Our forecasting pipeline leverages this structural understanding. The NNLS-blended ensemble of LightGBM, XGBoost, and Holt-Winters anchors the macroeconomic growth while reacting dynamically to holiday spikes, ultimately driven by engineered temporal features as verified by our SHAP explainability analysis.

While highly accurate for macroeconomic prediction, the pipeline lacks SKU-level granularity. The next step is decentralizing the global model into category-specific micro-models, allowing procurement to order Streetwear and GenZ inventory on completely isolated predictive timelines.

Code Availability

The full codebase, including data processing notebooks, the forecasting pipeline, and the generated dashboards, is publicly available at: https://github.com/quangtrinh25/vin_datathon.

References

- [1] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*, 30.
- [2] Chen, T., & Guestrin, C. (2016). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining* (pp. 785-794).
- [3] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30.
- [4] Winters, P. R. (1960). Forecasting sales by exponentially weighted moving averages. *Management science*, 6(3), 324-342.
- [5] Lawson, C. L., & Hanson, R. J. (1995). *Solving least squares problems*. Society for Industrial and Applied Mathematics.

A Appendix: Supplementary Dashboards

A.1 Dashboard 5: Product Analysis

This dashboard breaks down the performance and return rate of individual SKUs and overarching categories.

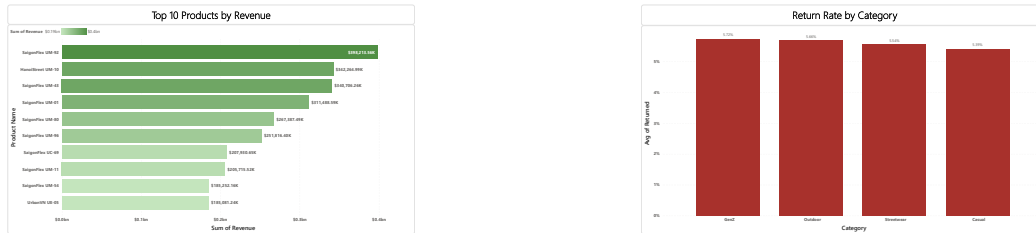


Figure 6: Hero SKUs dominate sales, but the GenZ category suffers from high return rates.

The “SaigonFlex UM-92” is the all-time best seller, generating \$398.2M alone. However, product quality control is a hidden tax on profitability; the overall return rate sits at 5.59%, with the GenZ category peaking at 5.72%. These “Hero” SKUs (Figure 6) must be placed on automatic reorder to prevent stockouts, but the company must simultaneously implement an automated quality-control alert—any SKU exceeding a 15% return rate for three consecutive months must be instantly pulled from the storefront for a sizing-chart review.

A.2 Dashboard 6: Operations & Logistics

This dashboard investigates supply chain efficiency, shipping delays, and checkout friction.

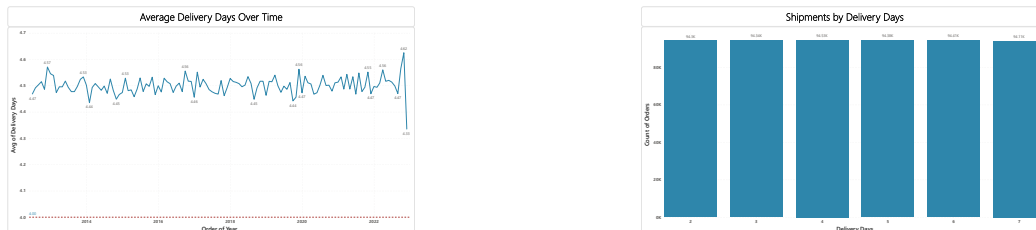


Figure 7: Uniform shipping delays and high checkout friction bleed \$1.5 Billion in intent.

The mean delivery time is 4.50 days, but the distribution is highly uniform, meaning the company treats all shipping equally rather than prioritizing based on cart value. More critically, the checkout cancellation rate is an alarming 9.19% (Figure 7), representing \$1,515.9M in lost top-line revenue over ten years. Operations must negotiate a strict 4-day delivery Service Level Agreement (SLA) with 3PL partners, and digital teams must conduct an immediate UX audit of the checkout flow—halving this 9.19% drop-off translates directly to a massive \$757.9M revenue recovery.

B Appendix B: Full Model Ablation Comparison

To validate our feature engineering and blending strategy, we conducted a rigorous ablation study. Table 2 details the isolated performance of each independent model before applying the final Non-Negative Least Squares (NNLS) weights and calibration factors.

Table 2: Ablation Study: Intermediate Model Performance

Model Architecture	Evaluate Score	Forecasted Revenue	Forecasted COGS
Organizing Committee Baseline	1,225,931.14	\$3,249,795	\$2,783,810
Holt-Winters Exponential Smoothing	—	\$3,058,695	\$2,770,203
XGBoost (Standalone)	—	\$3,176,207	\$2,767,520
LightGBM (Standalone)	—	\$3,266,419	\$2,842,544
NNLS Upgraded Ensemble	—	\$4,156,264	\$3,900,822
Final v57 MP Blend Ensemble	669,023.61	\$4,182,326	\$3,909,500